

Safeguard Effectiveness in Vibe Coding: An Empirical Evaluation of Test-Guided Repair and Regression in AI-Generated Full-Stack Web Applications

Can vibe-coded applications survive their own fixes? A deterministic, preregistered measurement of repair success, regression introduction, and their trajectories across repeated AI repair cycles — functional, security, accessibility, build, and persistence oracles included.

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This document freezes the study design, benchmark specifications, oracle suite, repair protocol, metric definitions, and analysis plan before data collection. No experimental results are reported. The accompanying repository implements every component and includes a harness-validation pilot whose data is excluded from all research claims.